Internals Know More Than the Model Can Say: Relevance Signals Are Encoded Internally and Outrun the Model's Own Readout

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Abstract

We study whether large language models (LLMs) "know" more about retrieval relevance than their outputs reveal. Using linear probes on a single frozen forward pass over (query, passage) pairs, we compare four relevance *readers* on the same model, prompt, and answer token: (i) the model's literal output verdict (zero-shot LLM-judge), (ii) a trained probe on the final-layer hidden state (the *output representation*), (iii) a trained probe on the best residual-stream layer (*internal*), and (iv) a probe on the top attention heads. Across eight models (3B–8B, LLaMA, Mistral and Qwen, base and instruct), reading internal activations decodes relevance markedly better than reading the output. With query-clustered bootstrap confidence intervals and Benjamini–Hochberg correction across the 8×4 family of tests, the attention-head probe beats the output-representation probe on all eight models—on every LLaMA and Mistral model and on both Qwen models, with corrected $q < 0.002$ throughout (effect +0.06 to +0.15). This edge survives a capacity-matched control that gives the final layer the identical read-out procedure, and the same internal activations are also far more decodable than the model's zero-shot verdict. We are explicit about the boundary: re-reading the final layer with a different read-out (a per-chunk ensemble) matches or beats the head probe on two of eight models, so the matched-procedure comparison and the margin over the model's own verdict are the claims we stand behind, not a general "internals beat the output end". The residual-stream advantage over the output representation is narrower: it is individually significant on the three *base* models (+0.078, +0.064, +0.059) and shrinks on their instruct counterparts, indicating that alignment moves relevance signal toward the output without closing the gap. Comparing base and instruct pairs, alignment reliably improves the output readout (the judge gets stronger), and the model's own readout direction still extracts less than a probe trained on the same layer on every model. We tried three routes to a mechanism and report all three, including where they fail. For the *residual-stream* reader a gradient control that edits nothing shows the verdict is not locally sensitive to the direction the probe reads relevance from, uniformly across the panel. For the *attention-head* reader—the one carrying our strongest effect—the same control is mixed, and a direct causal intervention on the probe direction finds no effect. In short: relevance is encoded internally, the model's own readout extracts only part of it, and we cannot say why.

1 Introduction

When an LLM is asked whether a retrieved passage is relevant to a query, its literal answer is often little better than chance. Yet the same forward pass may already contain a far cleaner relevance signal in its internal activations. We ask a focused, mechanistic question: *is relevance*

better decodable from a model's internals than from its output, and if so, is the internal signal causally connected to the output the model fails to produce?

This is an interpretability question, not a retrieval-engineering one. We do **not** claim a probe that beats a strong cross-encoder reranker, nor zero-shot cross-domain robustness; prior exploration refuted those framings. Our contribution is the *relative* science of internals vs. output:

- **Existence.** On the same model/prompt/answer-token, a linear probe on attention-head activations decodes (query, passage) relevance substantially better than the model's own output verdict, and better than a trained probe on the output representation (final hidden state).
- **Attribution.** Decomposing the gap into a *training* component (final-layer probe − judge) and a *depth/internal* component (best internal layer − final layer) rules out the obvious explanation and locates the rest: the training component is 0/8 after correction—fitting a probe on the output representation buys nothing over the zero-shot verdict—while the depth component is individually significant on the three base models and is compressed by alignment.
- **Mechanism.** Relevance decodability peaks in the middle of the network and declines toward the output layer. Comparing base vs. instruct models, alignment *improves* the output readout but does not eliminate the internal–output gap.
- **Causality.** We probe the causal role of the internal direction with a steering intervention whose verdict we pre-committed to: the internal direction has to produce a relevance-*aligned* shift that lands in the tail of a random-direction null distribution. We report the outcome in full, including the arm where it does not.

We report query-clustered bootstrap confidence intervals throughout, apply Benjamini–Hochberg correction across the family of edges, and are explicit about which claims are individually significant, which are null, and which hold on only a subset of models. Delineating effect strength honestly is central to this paper.

2 Related Work

Probing classifiers read latent features from frozen representations to test what information is linearly available [Alain and Bengio, 2017, Hewitt and Manning, 2019]; we use them comparatively across depth and against the output. **Logit lens and tuned lens** [nostalgebraist, 2020, Belrose et al., 2023] project intermediate states to the vocabulary, motivating our depth analysis. **Activation steering / representation engineering** [Turner et al., 2023, Zou et al., 2023, Li et al., 2023] edit residual directions to change behavior; we use a difference-of-means steering vector for a causal test. **Calibration and hallucination** work shows models can encode correctness they do not surface [Kadavath et al., 2022, Azaria and Mitchell, 2023, Burns et al., 2022]. Most relevant, Orgad et al. [2024] show LLMs internally represent truthfulness they do not output, and concurrent work finds hallucination is linearly decodable from mid-layer hidden states while output-based detectors stay near chance [Aiersilan, 2026]. We do *not* claim to be first to observe that internals outstrip the output; our contribution is to localize this to a specific, controllable signal—query–passage *relevance*—and to characterize it quantitatively: a depth-resolved decodability curve, a decomposition separating "reading internals" from "being trained," a read-out-matched capacity

control, a pre-registered causal steering test, and a base-vs-instruct contrast. **Relevance mechanism** work reverse-engineers how neural rankers compute relevance via causal interventions on attention heads [Chen et al., 2024]; trained classifiers are also known to beat prompted LLM judges at relevance [Gienapp et al., 2025]. We differ by studying a generative LLM's internal-vs-output gap on relevance rather than a ranker's internal circuit, and by attributing the gap to depth beyond training. **RAG relevance** estimation usually targets reranking accuracy; we deliberately stay on the interpretability question and do not compete with rerankers.

3 Method

Task and data. We use MS MARCO v1.1, forming (query, passage, label) triples with relevance labels. Splits are made *by query* (70/15/15) so no query appears in both train and test, preventing leakage. We report ROC-AUC on the held-out test set.

Prompt and extraction. Every reader sees the identical instruct prompt, `"Document: {p}\nQuestion: {q}\nIs the document relevant to the question? Answer:"`, and all activations are taken at the final *answer token*. Per-head activations are the `o_proj` outputs reshaped to $[\text{heads}, d_{\text{head}}]$; the residual stream is each decoder layer's output.

Four readers. (1) *LLM-judge*: the model's own normalized $P(\text{yes})$ vs. $P(\text{no})$ at the answer token, *zero training*. (2) *Output-representation probe*: an ℓ_2-regularized logistic regression on the final hidden state. (3) *Internal probe*: the same probe on the best residual-stream layer, selected *on validation*. (4) *Attention-head probe*: top-k heads chosen on validation, their activations concatenated into one logistic regression. As an untrained output-side control we also use a *logit lens*: each layer's residual is projected through the model's own final norm and unembedding to read $P(\text{yes})$ vs. $P(\text{no})$ at every depth, with no training and no label-based layer selection.

Gap decomposition. We split the total edge over the judge into train_edge = final − judge (pure training, both at the output) and internal_edge = best_resid − final (pure depth, beyond training). The thesis "internals > output" requires internal readers to exceed *both* the judge and the output-representation probe. A third component is read-out rather than depth: how the final layer is read. We therefore also hold the read-out procedure fixed (same feature budget, same validation-selected units, same per-unit standardisation) and, separately, replace the single linear read-out with a per-chunk ensemble, to separate "which layer" from "how it is read".

Significance. Because the split is by query, the correct resampling unit is the query. We report 95% confidence intervals and one-sided p-values from a *query-clustered paired bootstrap* (5000 resamples) on frozen test predictions; the best layer is chosen on validation to avoid selection-on-test optimism. Because we test four edges over eight models, we additionally report Benjamini–Hochberg q-values within each edge family (8 tests) and run a cross-model sign test on the direction of each edge.

Causal steering. At the validation-selected internal layer we form a probe-free relevance direction $d = \overline{h}_{\text{rel}} - \overline{h}_{\text{irrel}}$ (difference of class means, unit-normalized). During the forward pass we add $\alpha\|h\|d$ to the *output-layer* residual at the answer token and sweep α, reading the judge's $P(\text{yes})$ and the test AUC. Because a fixed step in α is not a fixed intervention—directions differ in how strongly they couple to the readout—we calibrate α per direction so that each sweep point corresponds to the same *shift in the yes/no logit difference*, and we report both the intended and the measured shift. This calibration has one boundary worth stating: when a direction's coupling to the readout is very weak, matching a large logit shift drives α into a numerically unstable range (we observe one random direction whose required α reaches ± 1266, where the logits overflow). Such points are dropped and the number of usable random directions is reported per endpoint (11 of

Table 1: Four relevance readers across eight models (test AUC, point estimates; val-selected layers and heads; a single regularisation strength, C=1.0, for all four readers so the columns are mutually consistent). The attention-head probe is the best reader on all eight models; the best residual layer also exceeds the output-representation probe on all eight. The zero-shot judge ranges from 0.545 (LLaMA-3.2-3B) to 0.660 (Qwen2.5-7B-Instruct).

Model	L	judge	final	best-resid	attn
LLaMA-3.2-3B (base)	28	0.545	0.536	0.613 (L9)	**0.684**
LLaMA-3.1-8B (base)	32	0.574	0.562	0.621 (L8)	**0.677**
Mistral-7B-v0.3 (base)	32	0.579	0.583	0.606 (L16)	**0.684**
LLaMA-3.2-3B-Instruct	28	0.614	0.612	0.638 (L18)	**0.712**
LLaMA-3.1-8B-Instruct	32	0.630	0.596	0.609 (L12)	**0.725**
Mistral-7B-Instruct-v0.3	32	0.646	0.596	0.634 (L13)	**0.713**
Qwen2.5-7B (base)	28	0.652	0.592	0.655 (L22)	**0.680**
Qwen2.5-7B-Instruct	28	0.660	0.610	0.630 (L23)	**0.669**

12 in that case). To ask whether steering is relevance-*aligned* rather than a generic bias knob, we track the *alignment gap* $\overline{P(\text{yes})}_{\text{rel}} - \overline{P(\text{yes})}_{\text{irrel}}$ separately over true-positive and true-negative rows: a relevance-bearing direction widens the gap, whereas a blunt shift moves both groups together. Each direction is then compared against a null distribution of twelve random unit directions (norm-matched, same calibration), so "the internal direction works" means it must land in the tail of that distribution—not merely that it moves the mean. The comparison set is the internal direction, the output layer's own difference-of-means direction, and the random directions.

4 Experiments

Internals decode relevance better than the output. Table 1 reports the four readers across eight models. The attention-head probe is the strongest of the four on *every* model, and the best internal reader exceeds the trained output-representation probe everywhere. The judge is weakest on the LLaMA and Mistral base models (AUC 0.545–0.579) and stronger on their instruct counterparts (0.614–0.646) and on Qwen (0.652–0.660); even where the judge is strongest, reading the heads still beats reading the output representation.

Significance. Table 2 gives query-clustered bootstrap CIs (5000 resamples, resampling *queries*) and one-sided p-values, with Benjamini–Hochberg q-values computed across the 8×4 family of edges (one family per edge, as is standard). The *attention-head vs. output-representation* edge is the only one that survives correction on all eight models; the residual-stream edge survives on the three base models; the "pure training" edge (final − judge) survives on none. This ordering is what the paper's claims rest on: the defensible headline is a *readout* comparison at matched training, and the residual-stream advantage is a base-model phenomenon that alignment compresses.

How the output representation is read matters as much as where. The headline edge compares a probe on 20 concatenated attention heads against a probe on the final residual vector. To separate "reading internals" from "reading differently" we re-read the *final layer* with the identical procedure: the same 2560-dimensional budget, the same validation-selected 20 units, the same per-unit standardisation and concatenation. The edge barely moves (+0.043 to +0.128, significant on 8/8 models), so the advantage is not an artefact of dimensionality or of the selection step. But the reverse control is equally informative: replacing the single linear read-out with a per-chunk logistic-regression ensemble raises the final layer from 0.536 to 0.621 on LLaMA-3.2-

Table 2: Query-clustered paired bootstrap (5000 resamples) at the 1500-query scale, all eight models. q is Benjamini–Hochberg within each edge family (8 tests). **Bold**: $q < 0.05$. attn−final survives on 8/8 models; internal_edge on the three base models; train_edge on none.

Model	internal_edge	attn−final	best_resid−judge	train_edge
LLaMA-3.2-3B (base)	+0.078 [.04,.12]	+0.148 [.11,.19]	+0.068 [.02,.11]	−0.009 [-.05,.03]
LLaMA-3.1-8B (base)	+0.059 [.02,.10]	+0.114 [.08,.15]	+0.047 [.00,.09]	−0.012 [-.05,.03]
Mistral-7B-v0.3 (base)	+0.023 [-.02,.06]	+0.101 [.07,.13]	+0.026 [-.02,.07]	+0.004 [-.03,.04]
LLaMA-3.2-3B-Instruct	+0.026 [-.01,.07]	+0.100 [.06,.13]	+0.024 [-.02,.07]	−0.002 [-.04,.04]
LLaMA-3.1-8B-Instruct	+0.013 [-.03,.06]	+0.129 [.09,.17]	−0.022 [-.06,.02]	−0.035 [-.08,.01]
Mistral-7B-Instruct-v0.3	+0.037 [-.01,.09]	+0.117 [.08,.15]	−0.013 [-.06,.03]	−0.050 [-.09,-.01]
Qwen2.5-7B (base)	+0.064 [.02,.10]	+0.088 [.05,.13]	+0.003 [-.04,.04]	−0.060 [-.10,-.03]
Qwen2.5-7B-Instruct	+0.020 [-.01,.05]	+0.059 [.02,.10]	−0.031 [-.07,.01]	−0.051 [-.09,-.01]

3B—a gain of +0.085, the same order as the entire internal_edge (+0.078). On two models the ensemble read-out of the final layer *matches or beats* the head probe outright (Qwen2.5-7B 0.683 vs. 0.680; Qwen2.5-7B-Instruct 0.719 vs. 0.669, i.e. a gap of −0.050). We therefore do not claim that internal readers beat the output end in general. The defensible claim, and the one the rest of the paper quantifies, is narrower: *at a matched read-out procedure* the internal layers yield the better relevance decoder, and against the model's own zero-shot verdict the internal advantage is large on every model. Both the positive control (matched procedure: 8/8) and the negative one (a different read-out of the same layer reverses the gap on 2/8) are reported here so that neither is left for a reader to discover.

The base-to-instruct contrast. Across all four base→instruct pairs the same triple appears: the judge improves (the output "says" more: 0.545→0.614, 0.574→0.630, 0.579→0.646, 0.652→0.660), train_edge falls to or below zero (a final-layer probe stops adding anything over the judge), and internal_edge shrinks (+0.078→ + 0.026, +0.059→ + 0.013, +0.064→ + 0.020; Mistral moves the other way, +0.023→ + 0.037). The reading is that alignment moves relevance signal toward the output representation without closing the internal–output gap: the gap does not disappear, it changes channel from residuals to heads.

A third model family (Qwen2.5) confirms and extends the pattern. Qwen2.5-7B has a stronger zero-shot judge than any LLaMA or Mistral base model (0.652 vs. 0.545–0.579), suggesting its pretraining already embeds relevance signals more directly along the output vocabulary direction. Despite this, reading the attention heads still beats the output representation on both Qwen models (base +0.088, instruct +0.059, both $q < 0.002$). The internal_edge is significant on the base model (+0.064, q=0.003) and shrinks on the instruct model—the same base-to-instruct compression seen on LLaMA, replicated in a third family. The train_edge is *negative* on both Qwen models (−0.060, −0.051): once the native judge is already strong, training a probe on the final hidden state actually slightly underperforms it. This is the structural mirror of the LLaMA case—where alignment *moved* signal toward the output (judge rising from 0.545 to 0.614 on 3B), Qwen appears to have that signal *already present* in the output pathway from pretraining. The internal–output gap persists regardless.

Depth localization. Figure 1 shows relevance decodability rising into the mid network and declining toward the output layer, with the judge baseline sitting below the peak; the same shape holds on all eight models (per-model figures in the artifact bundle). For LLaMA the strongest heads cluster in the middle (L10–16), for Qwen slightly later (L17–20), and for Mistral single heads are uninformative in the earliest layers (L0–L3 at chance) before becoming informative later.

The model's own unembedding reads out less than a trained probe (logit lens).

One might object that mid-network decodability only reflects what a *trained* probe can extract, not what the model itself could surface. We test this with a logit lens: at each layer we project the cached residual through the model's own final norm and unembedding and read the normalized $P(\text{yes})$ vs. $P(\text{no})$—an untrained, output-side reader at every depth. As a sanity check the last-layer logit lens reproduces the judge (maximum absolute deviation 0.011 across the eight models).

Comparing the two readers *at the same layer*—so that neither gets a free choice of depth—the trained probe wins on all eight models, by 0.005 to 0.129 AUC. The size of that gap is not cleanly predicted by training regime: it is largest on two base models (-0.129 LLaMA-3.1-8B, -0.102 LLaMA-3.2-3B) but also on an instruct model (-0.117 Mistral-7B-Instruct), and smallest on an instruct model (-0.005 LLaMA-3.2-3B-Instruct) and a base model (-0.040 Qwen2.5-7B). What is robust is the direction, not the split. Two caveats bound what this shows. First, the comparison is not capacity-matched in the learning sense: the probe is fitted on 5218 training pairs while the logit lens uses a *fixed* direction, so a gap means the model's own readout direction is not optimised for this task—not that the information is unreadable by any fixed direction. Second, applying the final norm at an intermediate layer is itself out of distribution.

We can measure the first caveat rather than assert it, on the FiQA cache where the two can be compared on identical data at an identical layer. There, the model's own fixed direction scores AUC $= 0.453$ at $L9$ with a query-clustered 95% CI of $[0.417, 0.489]$—*entirely below chance* ($P(\text{AUC} \geq 0.5) = 0.005$)—while a direction fitted on a *different* dataset (MS MARCO) and carried over with no target-domain fitting at all scores 0.568, and one fitted in-domain scores 0.777. On that dataset a direction that has never seen FiQA beats the model's own readout, and the model's own readout is significantly worse than chance.

We are careful not to generalise this. The same measurement on MS MARCO gives the model's own readout 0.511 with CI $[0.475, 0.546]$—indistinguishable from chance—and a FiQA-fitted direction carried to MS MARCO scores only 0.482, i.e. the transfer advantage *reverses*. So the below-chance result is a property of FiQA, not a universal statement about the model's readout. What does hold on both datasets, and is all we need, is the weaker comparison: the model's own readout direction extracts less relevance than a probe trained on the same layer, and this is a statement about the model's readout direction being off-task rather than about the information being unreadable by a fixed linear map.

What survives both caveats is the *relative* pattern: the same fixed readout loses far more at intermediate layers of base models than of aligned ones, which is consistent with alignment having moved relevance signal toward the output direction (Section 4).

Activation patching: the verdict does not follow the internal representation. The comparisons above establish that relevance is more decodable from internal layers. A stronger question is whether the model's own computation *uses* what is decoded there. We test this by replacing the answer-token activation at the best internal layer L^* with another test example's real activation, preferring a partner with the *opposite* label (all 1117 rows received an opposite-label partner), and asking whose label the verdict follows. Interpolating $h_{L^*} \leftarrow \alpha h_j + (1-\alpha) h_i$ keeps the manipulation in the residual stream's own coordinate system, so there is no cross-layer format change and no scale mismatch: both terms are in-distribution activations at the same layer.

Figure 2 shows the result across the panel. Three checks confirm the intervention is sound. Patching a sample with its *own* activation reproduces the natural verdict bit-for-bit ($\max |\Delta| = 0$) on every model, so the hook is inert when the payload is the identity. The freshly captured activations match the cache row-by-row (relative mean absolute error ≤ 0.004), ruling out a silent row misalignment. And the payload is not merely noise: replacing L^* with a *random* activation at matched norm stays near chance (0.499–0.520).

The result is model-dependent, and the split is informative. On both LLaMA base models the

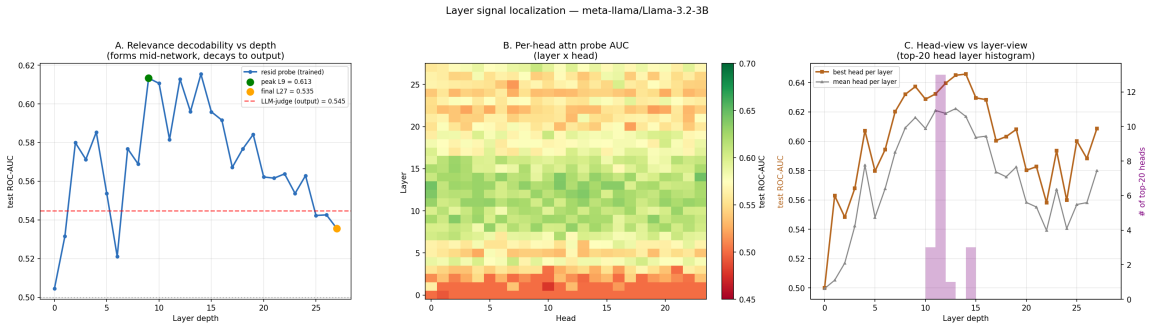


Figure 1: Depth localization (LLaMA-3.2-3B). (a) per-layer residual probe AUC vs. depth with the judge baseline and the peak/output markers; (b) per-head AUC heatmap; (c) head- vs. layer-view and the layer distribution of the top heads. Relevance decodability forms mid-network and is no longer rising at the output layer.

verdict does not follow the transplanted content at all: as α goes from 0 to 1 the AUC against the target label moves only $0.545 \to 0.534$ (3B) and $0.563 \to 0.550$ (8B), AUC against the *source* label only $0.455 \to 0.466$ and $0.437 \to 0.450$, and the relevant-minus-irrelevant gap in $P(\text{yes})$ changes by 0.003 and 0.006. A probe reads L^* at AUC $= 0.61$–0.62; the model's own forward pass from L^* onward extracts almost none of it. The instruct variant and Qwen2.5-7B behave differently, tracking the transplanted activation progressively (Δgap $+0.040$ and $+0.176$): on the instruct variant the target-label AUC falls from 0.607 to 0.392 while the source-label AUC rises from 0.393 to 0.608, and on Qwen the verdict ends up *anti*-correlated with the target label (AUC $= 0.337$) and correlated with the source label (0.663). So "the model cannot use its own internal representation" is false in general—these models' computations demonstrably can.

What distinguishes the models? We tested the obvious candidate—that tracking is weaker where the probe's margin over the output-side readers is larger—and found no support. On the eight models in Table 3 the Pearson correlation between Δgap and the probe's margin over the final-layer probe is -0.12 while the Spearman correlation is -0.48; the Pearson value was $+0.22$ before the last model was measured, so adding a single model flips its sign. Correlations with the probe's margin over the judge stay negative but wander over $[-0.69, -0.24]$ across leave-one-model-out subsets. An association whose sign depends on which models are included is not usable, so we report the patching pattern as model-dependent and leave its mechanism open rather than assert one.

A controller measurement establishes what *is* common across all eight models, and it is the sharpest test we have because it edits nothing. We compute the directional derivative of the model's own verdict with respect to h_{L^*}—a local property of the network *at the activation it actually produced*, so the out-of-distribution objection that limits patching does not apply—and take its cosine with the probe's unit relevance direction, comparing against sixteen random unit directions. Across all eight models the probe direction is never an outlier: the cosine is $+0.019$ at most and -0.010 at least, while the random-direction spread reaches magnitudes of 0.016 to 0.037, placing the probe direction between the 0.06 and 0.81 percentile of that null (Table 4). The output readout is therefore, to first order, *not pointed at the direction from which the residual-stream probe reads relevance*—and we stress *residual-stream* because, as the next paragraph shows, the head probe does not reproduce this. Crucially this holds for the Qwen models too, i.e. *even where the verdict is demonstrably driven by L^* content.* That is the misalignment, measured without perturbing anything: the information is present and the downstream network uses L^*, but the

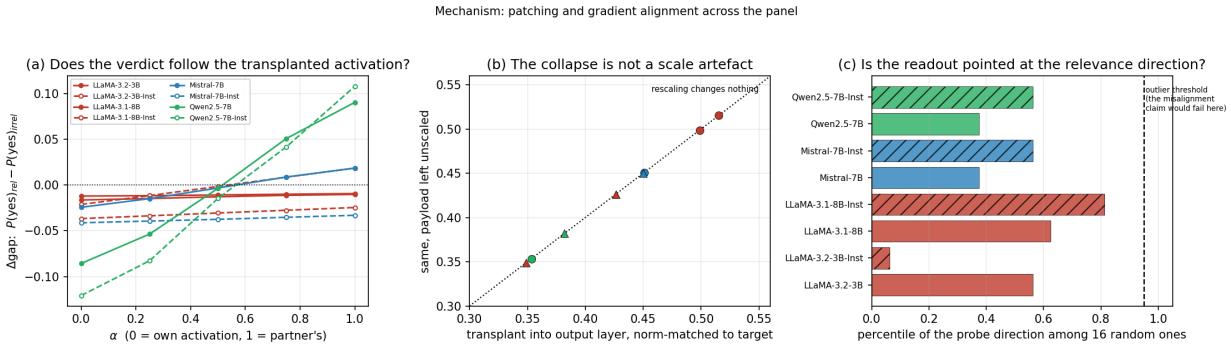


Figure 2: Activation patching across the panel. (a) The relevant-minus-irrelevant $P(\text{yes})$ gap as the internal activation is progressively replaced by an opposite-label example's. The two Qwen models track the transplanted label strongly (and at α=1 are anti-correlated with their own), the two LLaMA base models are flat, and the instruct variants sit in between—so "the model cannot use its own internal representation" is false in general. (b) Transplanting the same mid-layer activation into the *output* layer, with the payload rescaled to that layer's own norm versus left unscaled: every model sits on the diagonal, so the failure to transfer is not a scale artefact. (c) The gradient-alignment percentile from Figure 3; the dashed line is the value at which the misalignment claim would fail.

direction the verdict is locally sensitive to is not the relevance direction.

The same test on the reader that carries the headline. The comparison above is taken on the *residual stream*, but the paper's strongest effect is the *attention-head* probe. We therefore repeated the measurement with the gradient taken with respect to the selected heads' o_{proj} outputs, across the (two to four) layers those heads occupy, and projected into the probe's own coordinate system by dividing each head's gradient by that head's training standard deviation—the chain rule through the probe's standardisation. Panel (b) of Figure 3 shows the result, and it is *not* a uniform replication of panel (a). With 20,000 random directions the head-probe direction reaches the 0.99 and 0.98 percentile on LLaMA-3.1-8B-Instruct and LLaMA-3.1-8B (permutation $p = 0.006$ and 0.012), and 0.90/0.89 on Qwen2.5-7B and Mistral-7B ($p = 0.048/0.055$); but it sits at the 0.13–0.83 percentile on the remaining four models and is negative on three. Five of eight models have a positive mean cosine, which a sign test does not distinguish from chance ($p = 0.36$).

We read this as a genuine difference between the two readers rather than as noise, and it changes what the paper may claim. For the residual-stream probe the verdict is never locally aligned with the decoded relevance direction—that result is uniform across the panel and is what we call misalignment. For the head probe the picture is mixed: on the two LLaMA-3.1-8B models the verdict *is* measurably sensitive to the direction the probe reads, which suggests the head probe's advantage may partly reflect alignment rather than only decodability. We do not have a mechanistic account of why the two readers differ, and we therefore restrict the misalignment claim to the residual stream and report the head result as an open thread rather than extending the story.

A causal test on that direction finds nothing. Gradient alignment is a local, first-order statement; the stronger question is whether the verdict *depends* on the direction the head probe reads. We test that directly and it comes out negative, which is why this paper does not claim a mechanism for its headline effect.

The gradient analysis localises the leverage: for every model, one *dominant layer* carries 45–76% of the total gradient magnitude over the selected heads (LLaMA-3.2-3B L11 at 0.64, LLaMA-3.1-8B

Table 3: Cross-sample patching: does the verdict follow the transplanted activation? Δgap is the change in the relevant-minus-irrelevant $P(\text{yes})$ gap from α=0 to α=1. The middle columns test the candidate explanation that tracking is weaker where the probe leads the output-side readers more; no usable pattern survives. The last column compares transplanting the mid-layer activation into the *output* layer with and without rescaling it to that layer's own norm: the two agree to ≤ 0.0002 on all eight models, so the failure to transfer is not a scale artefact.

Model	probe@L^*	final	judge	probe−final	Δgap	norm vs. raw
Qwen2.5-7B-Instruct	0.630	0.610	0.660	+0.020	**+0.228**	0.382 / 0.382
Qwen2.5-7B	0.655	0.592	0.652	+0.063	+0.176	0.353 / 0.353
Mistral-7B-v0.3	0.606	0.583	0.579	+0.023	+0.043	0.451 / 0.451
LLaMA-3.2-3B-Instruct	0.638	0.612	0.614	+0.026	+0.040	0.349 / 0.349
LLaMA-3.1-8B-Instruct	0.609	0.596	0.630	+0.013	+0.012	0.426 / 0.426
Mistral-7B-Instruct	0.633	0.596	0.646	+0.037	+0.008	0.450 / 0.450
LLaMA-3.1-8B	0.621	0.562	0.574	+0.059	+0.006	0.515 / 0.515
LLaMA-3.2-3B	0.613	0.536	0.545	+0.077	+0.003	0.499 / 0.499

L15 at 0.48, Qwen2.5-7B L18 at 0.59), so a causal intervention has a well-defined target rather than being smeared over twenty heads in four layers. We inject the probe direction into those heads' o_{proj} outputs, calibrate each direction to the same yes/no logit shift, and measure the label-grouped alignment gap against random directions drawn from the *same* head subspace (the protocol of §4, including its saturation guards).

On LLaMA-3.2-3B, with 24 random directions and 512 test rows, the probe's gap is -0.0083 at a -2 shift (95% CI $[-0.035, +0.018]$) and $+0.0096$ at $+2$ ($[-0.006, +0.026]$): both intervals contain zero, and the permutation p-values against the pooled null of random directions are 0.44 and 0.20. The probe's gap does flip sign with the shift direction, so it is not pure noise, but its magnitude is not distinguishable from the random directions—at the -2 shift the largest random gap is -0.017, more than twice the probe's. Running the same test at two other settings we found that *which* shift passes the 95% criterion changes: at 12 directions and 1117 rows the negative shift passed and the positive failed; at 48 directions and 512 rows the reverse. A criterion satisfied on one side or the other depending on the configuration is not evidence of a causal effect.

So our three attempts at mechanism are consistent with one another and jointly negative. Patching shows the verdict tracks transplanted content on some models and not others, with no monotone relation to the size of the head advantage. Gradient alignment is mixed: two of eight models align significantly, three are negative, and the cross-model sign test is not significant. And the direct causal intervention finds no effect. What survives is a clean descriptive result—the attention heads are the best place to read relevance, on all eight models—and no account of why the model's own readout fails to use them. We state that as the limitation it is rather than dressing up a null as a mechanism.

For comparison, transplanting the same real mid-layer activation into the *output* layer also fails to transfer discriminability, and—this is the point of the control—*norm-matching the payload to the target layer's own norm does not rescue it*: at α=1 the norm-matched payload gives AUC = 0.499 and the unmatched one 0.499 on LLaMA-3.2-3B, 0.515 versus 0.515 on LLaMA-3.1-8B, and 0.353 versus 0.353 on Qwen. An earlier version of this experiment omitted that control, and the observed collapse could have been an artefact of injecting a vector an order of magnitude too small; with the control in place, rescaled and unscaled injections are indistinguishable, so the collapse is not a scale artefact. (The earlier version patched each example with its *own* activation and therefore measured a different quantity; its point estimates differ by 0.01–0.02 from the cross-sample version

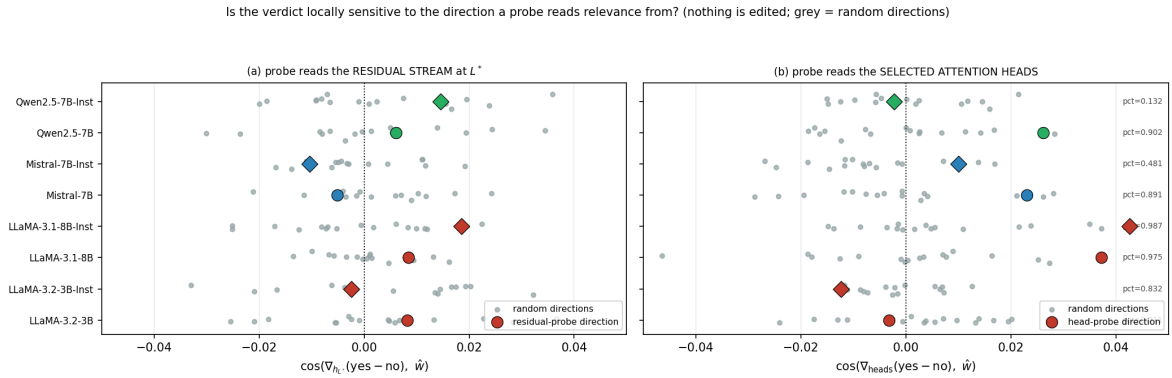


Figure 3: Gradient alignment, all eight models. For each model we compute the gradient of its own yes/no verdict with respect to h_{L^*} and take the cosine with the probe's relevance direction (coloured marker), comparing against sixteen random directions (grey). Nothing is edited, so no out-of-distribution concern arises. The probe direction is never an outlier—its percentile among the random directions ranges from 0.06 to 0.81, far from the 0.95 threshold—which is the misalignment result in its cleanest form.

Table 4: Gradient alignment, all eight models. cos is between the gradient of the yes/no logit difference with respect to h_{L^*} and the probe's unit relevance direction; the percentile places that cosine among sixteen random unit directions. Nothing is edited, so no out-of-distribution concern arises. The probe direction is never an outlier—its percentile at the 0.95 threshold would indicate an aligned readout—which is the misalignment result in its cleanest form.

Model	L^*	cos	random max |mean|	percentile
LLaMA-3.2-3B	9	+0.008	0.037	0.56
LLaMA-3.1-8B	8	+0.008	0.016	0.62
LLaMA-3.2-3B-Instr.	18	−0.002	0.033	0.06
LLaMA-3.1-8B-Instr.	12	+0.019	0.025	0.81
Mistral-7B-v0.3	16	−0.005	0.024	0.38
Mistral-7B-Instr.	13	−0.010	0.019	0.56
Qwen2.5-7B	22	+0.006	0.035	0.38
Qwen2.5-7B-Instr.	23	+0.015	0.036	0.56

here and the two should not be compared numerically.)

The honest caveat is that cross-example patching introduces activations the network never produced for *this* input, so part of any insensitivity could reflect the model ignoring out-of-distribution content rather than ignoring relevance specifically. That caveat does not apply to the gradient measurement above, which is why we rely on it for the misalignment claim and treat the patching as corroborating rather than primary evidence. It also cuts against over-reading the Qwen result: tracking a transplanted activation is not the same as using relevance, and Qwen's own gradient alignment is what would settle whether its readout is pointed at relevance.

Mechanism: base vs. instruct. Across all four base→instruct pairs, alignment shows a consistent triple effect: the judge improves (output "says" more), train_edge falls toward or below zero (training a final-layer probe adds little once aligned), and internal_edge *shrinks* in three of four pairs while the head edge persists. Combined with Table 2 (internal_edge significant on base, compressed on instruct), the reading is: *alignment moves relevance signal toward the output*

representation but does not close the internal–output gap—it does not merely "suppress" the output.

Causality: what steering does and does not show. We pre-committed to a directional criterion before running the intervention: for steering to count as *relevance-aligned*, the internal direction's alignment gap (the relevant-minus-irrelevant split in $P(\text{yes})$) had to land in the upper tail ($\geq$95th percentile) of a null distribution formed by twelve random directions swept over the same calibrated logit shifts. It does not. On LLaMA-3.2-3B the internal direction's gap sits at the 0.55 and 0.45 percentile at the two ends of the sweep, and on LLaMA-3.1-8B at the 0.33 and 0.25 percentile (Figure 4, panel c). Two independently trained base models therefore agree: at the registered operating point the internal direction is not a relevance-*aligned* intervention.

The criterion was registered at the sweep "endpoints" (the $\pm$8-logit shifts), and inspecting the full profile shows that this is a weak place to test: at those points every direction has been pushed hardest, and the alignment gap is compressed toward zero for *all* of them, so a direction whose effect lives at moderate strength is structurally invisible there. We do not move the criterion—post-hoc adjustment is exactly what pre-registration prevents—but we report the full profile, and it changes the picture for two of the four models. On Qwen2.5-7B, at a -2-logit shift the internal direction's gap (0.166) exceeds the maximum of all twelve random directions (0.143), i.e. the 1.00 percentile. On LLaMA-3.2-3B-Instruct the effect is broader rather than a single point: the gap roughly *triples* over baseline ($0.026 \rightarrow 0.083$) and lands at the 1.00 percentile at both -1 and -2-logit shifts. The two base LLaMA models, by contrast, never exceed the 0.75 and 0.42 percentile anywhere in the sweep.

So the honest summary has three parts. At the registered endpoint criterion, the result is a null on all four models. At moderate strength, two of four models (both instruct) show a relevance-aligned effect beyond the random-direction null, while both LLaMA base models do not—a split that runs in the same direction as the base/instruct split in internal_edge, though four models (one a single point, one with only five random directions) are too few to call it a finding. And the endpoint choice itself was a weakness of our registered criterion: separating directions works worst exactly where we chose to test. We report the criterion as registered, and the profile as measured. The saturation-free calibration is what makes these comparisons interpretable at all: the three direction families are comparable after calibration (mean logit shift per unit α on 3B: internal -0.56, final $+3.27$, random -1.24; i.e. roughly a sixfold spread that the per-direction calibration removes).

The sweep also exposes *why* the gap fails to grow, which is more informative than the null itself: the alignment gap shrinks monotonically with intervention strength for *every* direction, including the random ones (on 3B the mean random gap falls from 0.0121 at baseline to 0.0016 at the strongest negative shift; on 8B from 0.0164 to 0.0012). Broadly perturbing the output residual compresses label separability rather than sharpening it, so the internal direction is a blunt verdict knob, not a relevance-aligned one.

What survives is the ablation arm. Pushing the yes/no logit difference negative along the internal direction collapses the verdict monotonically ($P(\text{yes})$ $0.814 \rightarrow 0.691 \rightarrow 0.573 \rightarrow 0.456 \rightarrow 0.393$ on 3B; $0.886 \rightarrow 0.260$ on 8B), and the internal direction is the only one of the three that does so *without* destroying discriminability: AUC stays in $[0.522, 0.546]$ on 3B and moves by 0.007 on 8B, whereas the output layer's own direction drives AUC down to 0.481 (3B) and 0.501 (8B), and the random directions move the mean without a label-aligned component. So the internal direction is causally coupled to the verdict, but we cannot show that it is causally coupled to relevance *discrimination*. The correlational results above do not depend on this arm, and we prefer to bound them with a reported null than to rest them on a claim the data does not support.

A second in-domain dataset. To check the gap is not specific to MS MARCO, we retrain and re-test the readers in-domain on BeIR FiQA (LLaMA-3.2-3B, 1500 queries, 225 test queries,

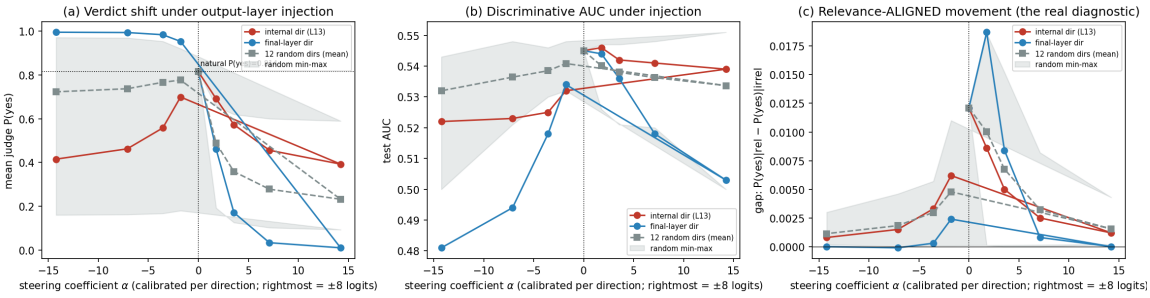


Figure 4: Steering at the output layer (LLaMA-3.2-3B), calibrated in logit-shift space so the three direction families are swept over the same yes/no logit differences rather than the same raw coefficient. (a) mean judge $P(\text{yes})$; (b) test AUC; (c) the alignment gap, with the random directions' mean and min–max band. All directions compress the gap as the intervention strengthens; the internal direction is not an outlier, which is the pre-registered null we report.

1125 test pairs; *not* cross-domain transfer). Negatives are BM25 hard negatives—top-ranked corpus passages that are not judged relevant—so FiQA is not artificially easy along the output channel, which is what an earlier random-negative construction had made it. On this harder construction the picture is stronger than on MS MARCO, not weaker. The judge reaches only AUC $= 0.597$ while the best internal layer reaches 0.829 (best_resid−judge $= +0.225$, CI $[0.188, 0.262]$), and the output representation no longer sits near a ceiling (0.724). The internal_edge is the largest in the paper ($+0.099$, CI $[0.066, 0.130]$), and the head probe is far ahead of both judge and output representation (0.884; attn−final $= +0.161$, CI $[0.131, 0.189]$). The two datasets are therefore complementary in the opposite direction from what one might expect: MS MARCO provides the harder *readout* comparison (a strong residual probe close to the output representation), while FiQA shows the internal advantage widening once the negatives are hard. **A stronger judge does not close the gap.** A reviewer might attribute the weak judge to a weak prompt rather than to the output pathway. Whatever the prompt, the judge remains the weakest of the four readers on every model we tested, and the zero-shot judge on the best-prompted large model (Qwen2.5-7B-Instruct, AUC $= 0.660$) is still below a linear probe on the same model's attention heads (0.669). Prompt engineering moves the judge; it does not put the judge ahead of the internal readers.

5 Discussion and Limitations

The relevance signal an LLM needs to judge a passage is present in its internals and only partially surfaced at the output. The results above support a specific reading of why, and it is a claim about *misalignment* rather than about uniform weakening. Three measurements point the same way. First, the signal is *linearly decodable* at mid-network layers while the model's *own* readout direction extracts much less from the *same* layer: at $L9$ of LLaMA-3.2-3B a trained direction reaches 0.61 on MS MARCO and 0.78 on FiQA, whereas the model's own direction reaches 0.51 and 0.45 respectively (and on FiQA that 0.45 is significantly below chance). Second, activation patching shows this is not a matter of format incompatibility: a mid-layer representation transplanted into the output layer transfers no discriminability, and rescaling it to the output layer's own norm does not help, so the failure is not that the vector arrives too small. Third—and this is the sharpest test—the model's verdict is, to first order, *not locally sensitive to the direction a probe reads relevance from*: the cosine between the gradient of the verdict with respect to h_{L^*} and the probe's direction is 0.008 (LLaMA-3.2-3B) and 0.006 (Qwen2.5-7B), against a random-direction spread

reaching |0.037| and |0.035|, placing the probe direction at the 0.56 and 0.38 percentile of that null. Crucially this holds *even for the model whose verdict is demonstrably driven by L^* content*, which is why we read the mechanism as the readout being pointed elsewhere rather than as the downstream network being unable to use the representation. Alignment then moves the readout toward the signal without closing the gap. This connects to calibration/hallucination findings that models "know" more than they assert, localized here to a specific signal and resolved over depth. The Qwen2.5 family is instructive: its zero-shot judge is already strong, so the internals-vs-*judge* gap nearly closes, yet reading attention heads still beats reading the output representation—the internal–output gap is robust to how well the output verbalizes relevance.

Limitations. (0) The paper's headline effect has no mechanism that we could detect, and we consider this its most important weakness. The residual-stream *misalignment* result is solid but is a result about a *different, weaker* reader: the residual probe beats the output representation on only three of eight models, so it is not the effect this paper is built on. For the attention-head reader, which beats the output representation on all eight, all three of our mechanistic probes come back negative or mixed—patching is model-dependent with no relation to the size of the advantage, gradient alignment is significant on two of eight models and non-significant across the panel, and the direct causal intervention on the probe direction has confidence intervals containing zero at both shift targets ($p = 0.44$ and 0.20 against the pooled null). We therefore present a descriptive result and an honest account of the failed attempts. A reader who wants the mechanism behind "the heads decode relevance better than the output" will not find it here. (1) Probing is correlational, and our causal evidence is weaker than our correlational evidence; the steering and activation-patching controls are reported in full, including where they fail to separate competing explanations (see the causality experiments below). (2) We use two in-domain relevance datasets (MS MARCO, FiQA); we make no cross-domain or reranking-accuracy claims, which prior exploration refuted. (3) The headline edge is defined against one specific output-side reader: a single C=1.0 logistic regression on the final residual vector. Re-reading that same layer with a per-chunk ensemble closes or reverses the edge on two of eight models (see the read-out control above), so "internal readers beat the output end" is *not* what we claim; what holds is the matched-read-out comparison plus a large margin over the model's own verdict. (4) Absolute AUC values are modest (0.54–0.73) on MS MARCO; the contribution is the *relative* internal-vs-output relationship, not a state-of-the-art relevance classifier. (5) The clean residual-stream comparison (internal_edge) is significant only on base models; on instruct models it is positive but not individually significant, so the strong form of the "residual stream knows more" claim is a base-model result.

6 Conclusion

On the same model, prompt, and token, relevance is more decodable from internal activations than from the output the model produces. The advantage is not explained by training alone, it is largest under a matched read-out procedure, and it holds in both directions of the base/instruct comparison: alignment improves what the model says without making its output the better place to read relevance from. The mechanism is best described as *misalignment* rather than attenuation: the information is present and the downstream network does use the internal layer, but the direction the verdict is locally sensitive to is not the relevance direction—a mismatch that replicates on every model for the residual-stream reader. That reader, however, is not the one carrying our strongest result. For the attention-head reader we tried three routes to a mechanism—patching, gradient alignment, and a direct causal intervention—and all three came back negative or mixed, so we make no mechanistic claim about the effect we are most confident in. What we can defend is the

measurement: across eight models, three families, and both training regimes, the attention heads are the best place to read whether a passage is relevant, and the model's own verdict is the worst. Internals know more than the model can say; why the model does not use what its heads encode remains open.

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